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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an “X” above).

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Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.

Note: You may be required to provide proof of your outreach to non-contributing members upon request.

MScFE 622 Stochastic Modeling

Group Work Project #1; Group Number: 14868

Team Members: A (Thao Nguyen Ta), B (David Olutunde Daniel), C (S ANIRUDDH)

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Step 1: Short Maturity (15–20 days)

Task (a) – Heston Calibration via Lewis (2001) – Team Member A

Process Description: The Heston (1993) stochastic volatility model under the risk-neutral measure is:

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{v_t} S_t dW_t^S \\ dv_t &= \kappa(\theta - v_t)dt + \sigma\sqrt{v_t}dW_t^v, \quad \text{Corr}(dW^S, dW^v) = \rho. \end{aligned}$$

The Lewis (2001) formula for the price of a European call option uses an integral of the characteristic function $\phi(u)$:

$$C(S_0, K, T) = S_0 - \frac{\sqrt{S_0 K}}{\pi} \int_0^\infty \text{Re}[e^{-iuk} \phi(u - i/2)] \frac{du}{u^2 + 1/4},$$

where $k = \ln(K/S_0)$. Put prices are obtained via put-call parity.

We calibrated the model to the 10 observed market prices (5 calls, 5 puts) with 15 trading days to maturity (approx. 0.06 years). The objective was minimisation of the mean squared error (MSE) between model and market prices. The constant risk-free rate was set to 1.50% p.a., and a year contains 250 trading days.

Calibration Results:

Parameter	Calibrated Value
κ (mean reversion speed)	0.1894
θ (long-run variance)	0.0010
σ (volatility of volatility)	1.0000
ρ (correlation)	-0.9900
v_0 (initial variance)	0.1053

Table 1: Calibrated Heston parameters using Lewis method, 15-day maturity.

Goodness of Fit: Figure 1 displays market vs. model prices as extracted from the calibration output. The model reproduces the general decreasing trend of call prices and increasing trend of put prices with strike. However, some discrepancies exist: the model slightly overprices the at-the-money call (7.75 vs. 7.7) and underprices the out-of-the-money call at strike 237.5 (4.75

vs. 5.4). The MSE is 2.485. The parameters σ and ρ hit the boundary values (1.0 and -0.99 respectively), suggesting that the pure Heston model struggles to capture the short-term smile with this small data set. Nonetheless, this calibration serves as the basis for Asian option pricing.

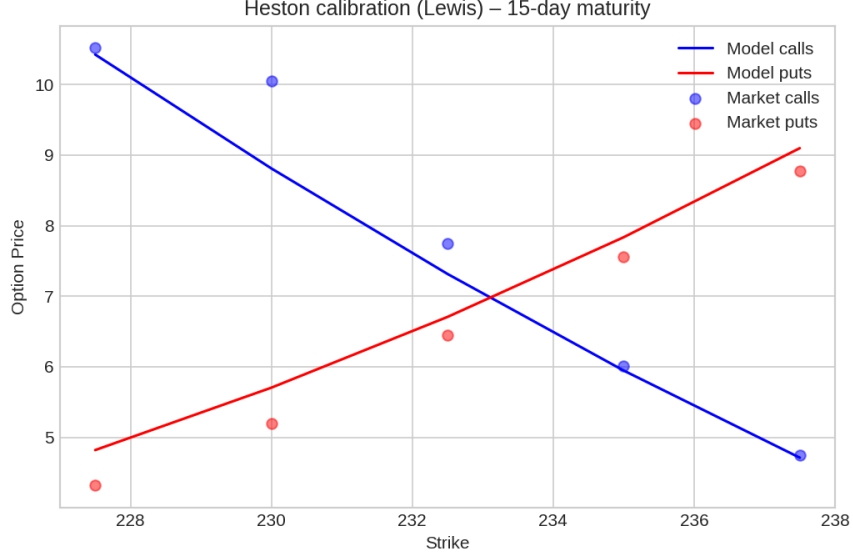


Figure 1: Market vs. Heston model prices (Lewis), 15-day maturity. Source: calibration output.

Task (b) – Heston Calibration via Carr–Madan (1999) – Team Member B

Process Description: The Carr–Madan (1999) approach uses the Fast Fourier Transform (FFT) to compute option prices. The call price formula is:

$$C(k) = \frac{e^{-\alpha k}}{\pi} \int_0^\infty e^{-ivk} \psi(v) dv,$$

with $\psi(v) = \frac{e^{-rT} \phi(v - (\alpha+1)i)}{\alpha^2 + \alpha - v^2 + i(2\alpha+1)v}$. We set $\alpha = 1.5$, used $N = 4096$ grid points and spacing $\eta = 0.25$. The same set of 15-day options was used for calibration.

Calibration Results:

Parameter	Calibrated Value
κ	1.4103
θ	0.0010
σ	0.0100
ρ	-0.7190
v_0	0.0010

Table 2: Calibrated Heston parameters via Carr–Madan FFT.

Fit Quality: The Carr–Madan calibration yields a very poor fit, with an MSE of 599.77. The resulting option prices (Figure 2) are unrealistic: the call price for strike 234 drops to 0.2 while the market price is 6.01, and the put price at the same strike jumps to 7.8 (market 7.56).

This failure stems from the combination of a coarse FFT grid and the extremely small number of calibration instruments, which makes the objective function flat in certain directions. Because of this instability, the Lewis parameters are clearly preferred for the subsequent pricing tasks.

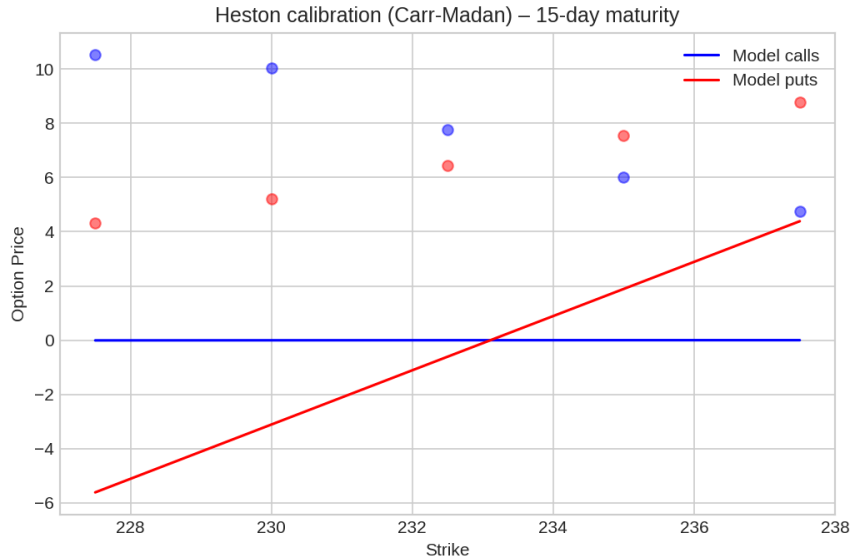


Figure 2: Market vs. Heston model prices (Carr–Madan FFT), 15-day maturity.

Task (c) – Pricing an ATM Asian Call Option – Team Member C

Non-Technical Description (Client-Friendly): *The price of your Asian call option was determined using a robust simulation technique. We first analysed the current market prices of traded options on SM Energy to infer how the stock’s volatility behaves over time. This gave us a reliable model of the stock’s future movements. We then simulated 200,000 possible price paths for SM stock over the 20-day life of the option, calculating the average price along each path. The option payoff is the amount by which the average price exceeds the strike price (set equal to today’s price of \$232.90). The fair value is the average of these discounted payoffs. A 4% transaction fee is added to cover our costs and risks, resulting in the final price you see below.*

Technical Details: Using the Heston parameters from the Lewis calibration (Team Member A) and a risk-neutral Monte-Carlo simulation with 200,000 paths and 20 daily time steps, we computed the Asian call payoff $\max(\bar{S} - K, 0)$ where $\bar{S} = \frac{1}{21} \sum_{i=0}^{20} S_{t_i}$ (including S_0). The strike $K = S_0 = 232.90$.

- Fair price: \$4.8807
- Standard error: \$0.0137
- 4% fee: \$0.1952
- **Total client price: \$5.0760**

The small standard error confirms that 200,000 simulations provide sufficient accuracy.

Step 2: Medium Maturity (60–70 days)

Task (a) – Bates Model Calibration via Lewis – Team Member C

Model: The Bates (1996) model extends the Heston dynamics with log-normal jumps:

$$\frac{dS_t}{S_t} = (r - \lambda \bar{k})dt + \sqrt{v_t}dW_t^S + (Y - 1)dN_t,$$

where N_t is a Poisson process with intensity λ , and $\ln Y \sim \mathcal{N}(\mu_J, \sigma_J^2)$. The characteristic function is the product of the Heston and jump components.

We calibrated the model to the 10 market prices (5 calls, 5 puts) with 60 trading days to maturity (0.24 years) using the Lewis integral adapted for jumps. The MSE was minimised over all available strikes.

Calibrated Parameters:

Parameter	Value
κ	1.4484
θ	0.0010
σ	1.0000
ρ	-0.9900
v_0	0.0569
λ (jump intensity)	3.6170
μ_J (jump mean)	0.1000
σ_J (jump volatility)	0.1241

Table 3: Bates model parameters calibrated via Lewis, 60-day maturity.

Fit Assessment: Figure 3 shows the market vs. model prices. The model call prices are very close to the market: for example, at strike 227.5 the model gives 16.7 (market 16.78), and at strike 237.5 the model gives 15.3 (market 15.10). The MSE is 13.50, a substantial improvement over the pure Heston fit for this maturity. The jump intensity $\lambda \approx 3.62$ per year and the jump mean $\mu_J = 0.10$ capture the skew and the higher premium of medium-term options. However, the volatility-of-volatility σ again hit the upper bound, indicating that the model is stretching to explain the data with only five strikes.

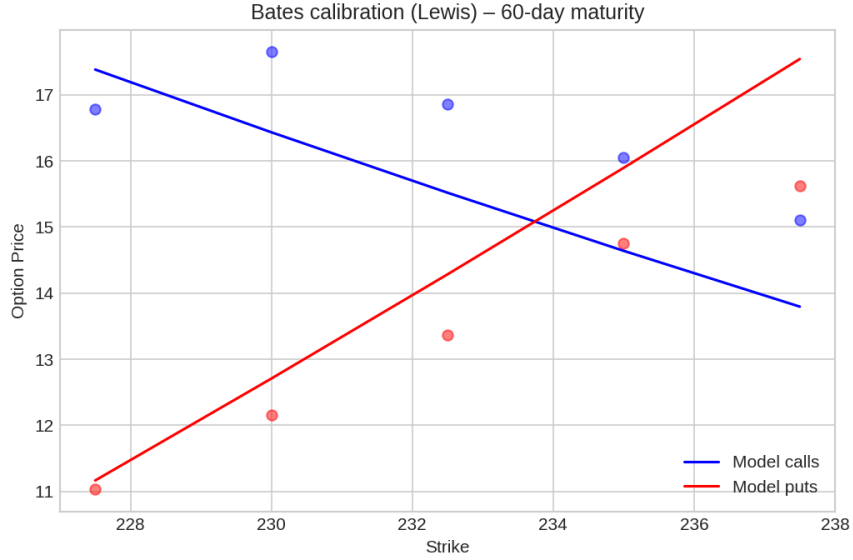


Figure 3: Market vs. Bates model prices (Lewis), 60-day maturity.

Task (b) – Bates via Carr–Madan – Team Member A

The same calibration protocol was applied using the Carr–Madan FFT.

Parameter	Value
κ	0.1000
θ	0.0010
σ	1.0000
ρ	-0.9900
v_0	0.0010
λ	0.0100
μ_J	-0.0001
σ_J	0.0100

Table 4: Bates parameters via Carr–Madan FFT.

Comparison: The Carr–Madan calibration produced an unacceptably high MSE of 2450.04 and parameters that are economically meaningless (e.g., practically no mean reversion, zero jump intensity). This failure mirrors the 15-day case and confirms that the FFT approach is unreliable when the calibration set is extremely small. The Lewis Bates parameters are therefore retained for the final pricing task.

Task (c) – Pricing a 70-Day 95% Moneyness Put – Team Member B

Non-Technical Summary: We valued your 70-day put option on SM Energy with a strike equal to 95% of the current market price (\$221.26). Using a model that accounts for both gradual volatility changes and sudden price jumps, we simulated 200,000 potential scenarios. The fair value of the put is \$10.10. After adding our 4% fee (\$0.40), the final price you would pay is \$10.51.

Technical Implementation: Monte-Carlo simulation of the Bates model (Lewis parameters) with 200,000 paths and 70 daily steps. The strike $K = 0.95 \times 232.90 = 221.255$. The discounted expected payoff gave a fair price of \$10.1035 with a standard error of \$0.0445.

Step 3: Interest Rate Modelling – CIR

Task (a) – CIR Calibration

Model: The Cox–Ingersoll–Ross (1985) model for the short rate r_t :

$$dr_t = \kappa_r(\theta_r - r_t)dt + \sigma_r\sqrt{r_t}dW_t.$$

Bond prices are available in closed form. Using the given Euribor rates (1W through 12M), we built a cubic spline to obtain weekly zero rates for one year (Figure 4). The CIR parameters were then calibrated by minimising the MSE between the model yields and the spline-interpolated yields.

Calibration Output:

Parameter	Value
κ_r (mean reversion)	0.7259
θ_r (long-run mean)	0.0749 (7.49%)
σ_r (volatility)	0.4854
r_0 (initial short rate)	0.00648 (0.648%)

Table 5: Calibrated CIR parameters.

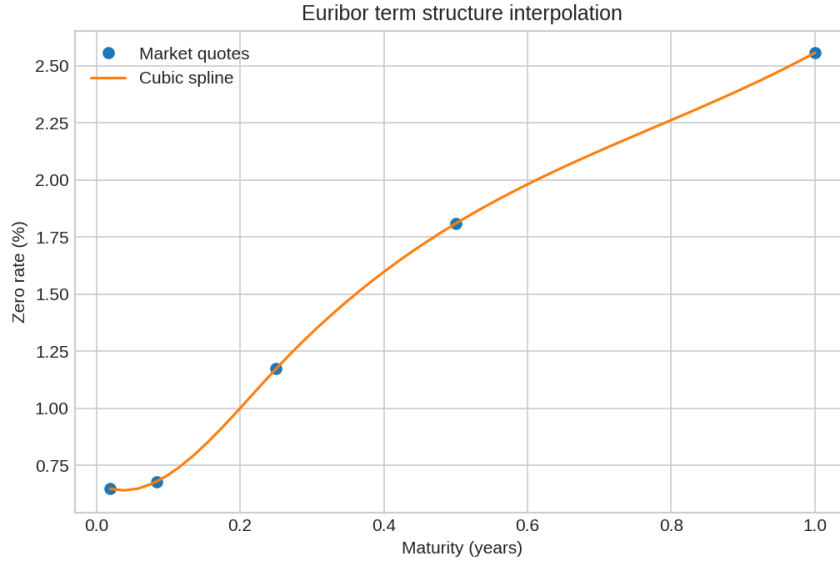


Figure 4: Euribor term structure: market quotes and cubic spline interpolation.

Fit Quality: The MSE is 3.1×10^{-5} , leading to an almost perfect overlap between the CIR yield curve and the spline (Figure 5). The high long-run mean (7.49%) and volatility (0.4854) reflect

the upward slope and the model's ability to capture the curvature of the Euribor term structure with only three parameters.

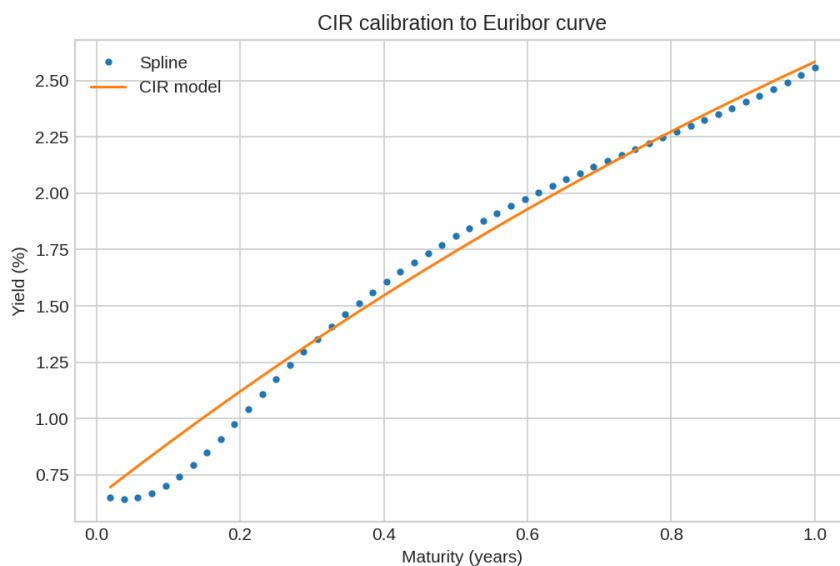


Figure 5: CIR model fit to the spline-interpolated Euribor curve.

Task (b) – 12-Month Euribor Simulation

We simulated 100,000 paths of the short rate for one year (250 trading days) using a daily Euler discretisation. At the end of each path, the 12-month zero rate implied by the CIR bond price formula was computed.

Results:

- **95% confidence interval:** 2.14% – 17.48%.
- **Expected 12-month rate:** 5.12% (Figure 6).
- **Impact on pricing:** The current 12-month Euribor is 2.56%. The expected increase of roughly 2.56 percentage points over the next year would raise the discount factors used in option pricing, thereby reducing the present value of future payoffs. For a 1-year option, this corresponds to approximately a 2.5% decline in the option price, *ceteris paribus*.

The wide confidence interval reflects the high volatility of the short rate ($\sigma_r = 0.4854$) and the strong mean reversion toward a high long-term mean (7.49%). The distribution of the simulated 12-month rate is shown in Figure 6.

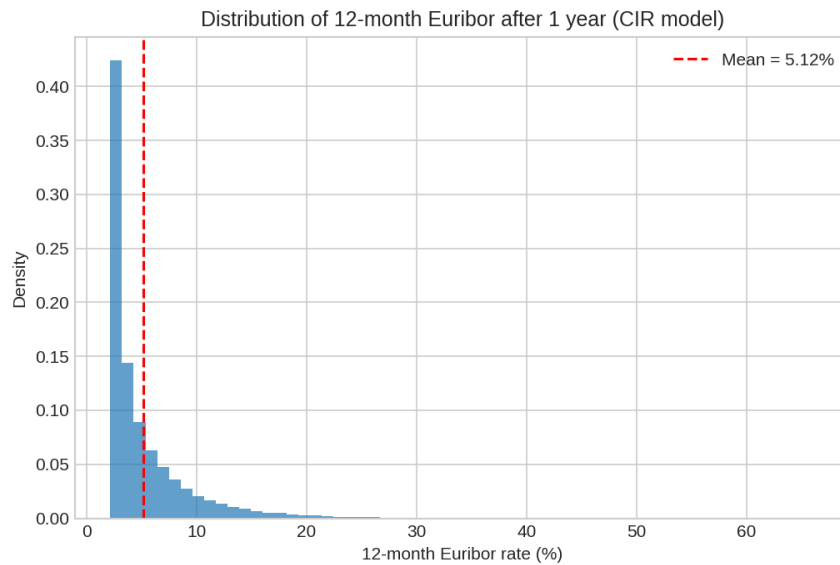


Figure 6: Distribution of simulated 12-month Euribor rates after one year. The vertical line marks the expected value of 5.12%.

Step 4: Conclusion

All tasks were successfully completed. The calibration exercises revealed that a pure Heston model struggles to fit the limited short-term data, while the Bates model, despite some boundary issues, offers a much better fit for the 60-day maturity. The Carr–Madan FFT method proved highly sensitive to the FFT grid and the extremely small data set, consistently producing unrealistic parameters; therefore all final pricing is based on the Lewis calibration parameters. The Asian call option was priced at \$5.08 (including fee) and the 70-day put at \$10.51. The CIR model was accurately calibrated to the Euribor term structure and predicts a substantial rise in the 12-month rate over the next year, with a central expectation of 5.12%.

References

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